UNIT 2 · TOPIC 6(A)

# Design of Seed Metering Systems

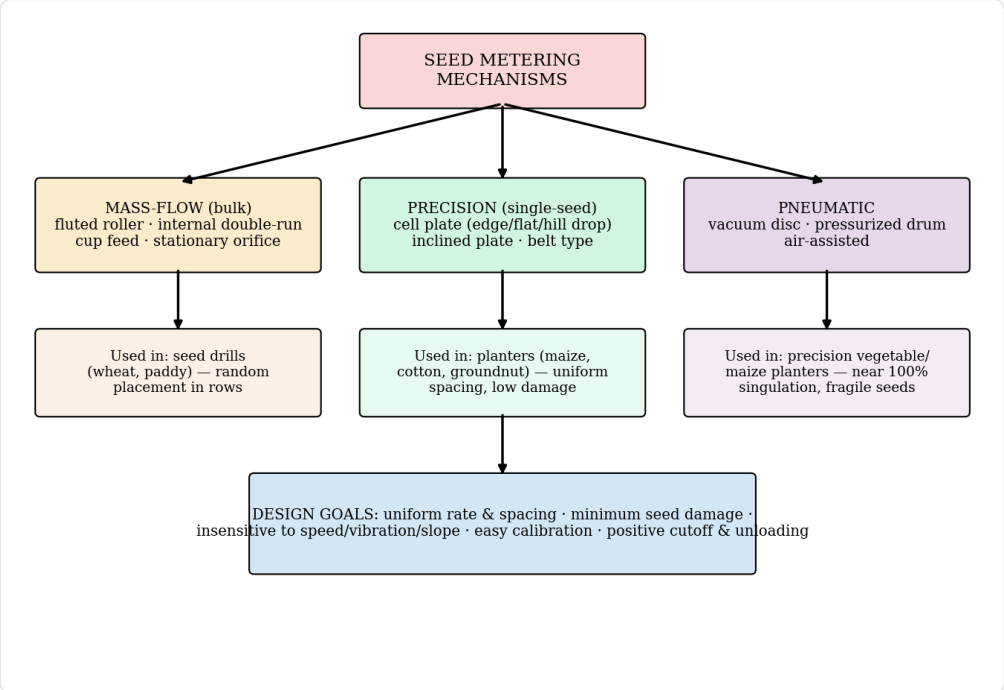
*Syllabus topic: Design of soil working tools for sowing and planting*

## Functional requirements of a sowing/planting machine

1. Meter seeds at the desired rate with uniform distribution/spacing.
2. Open a furrow to uniform depth.
3. Deposit the seed with minimum bounce/scatter.
4. Cover and firm the soil around the seed.
5. Cause minimum mechanical damage to the seed.

Kepner's requirements of a good precision metering unit (§11.16 area):

- Cells sized to the seed (one seed per cell — grading the seed helps).
- Proper cell fill (**95–100%** target).
- Gentle handling (low plate speed, smooth surfaces).
- A good **cutoff device** (prevents multiple fill and seed breakage).
- **Positive unloading** — ejection of seed from cells (knock-out devices).

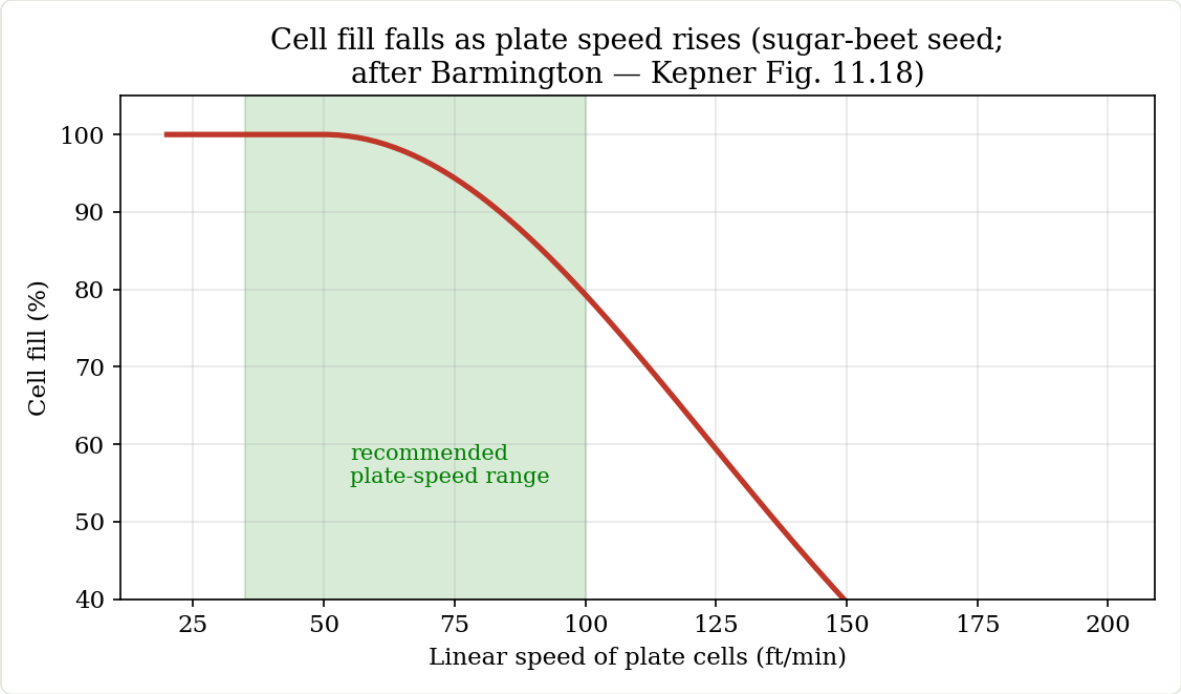



*Classification of seed metering mechanisms*

## Cell-plate (precision) metering — design variables

- **Cell size vs seed size:** make the cell slightly larger than the largest seed dimension. Graded/processed seed (e.g., segmented sugar-beet seed, delinted cotton) enables true singulation.
- **Plate (cell) linear speed** is the critical design limit — cell fill falls as plate speed rises. Test data (Barmington, in Kepner Fig. 11.18): **100% cell fill at ~50 fpm (~0.25 m/s)**; good range ≈ **35–100 fpm**. 95–100% cell fill also gives the most uniform spacing and minimum seed damage.
- 100% "fill" is statistical — empty cells are offset by multiple fills.

- For higher forward speeds, use **more cells per plate or a larger plate diameter — not a faster plate**.




*Cell fill vs plate speed*

$$s = \pi \cdot D_w \,/\, [n_c \times i \times (1 - \sigma)]$$

Seed spacing delivered by a planter with a ground-driven wheel: D_w = drive-wheel diameter, σ = drive-wheel skid (5–10%), n_c = cells per plate, i = plate-to-wheel speed ratio. With no skid this is the standard s = πD/(N·C) of the ICAR eCourse (Farm Machinery & Equipment I, Lesson 24).

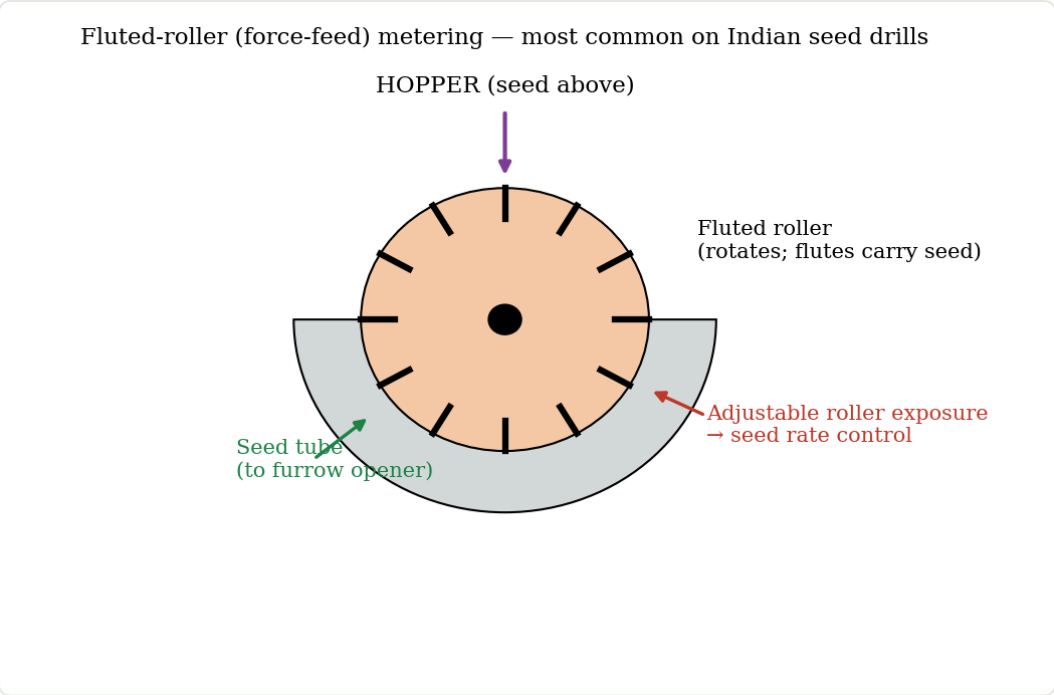
**Corrected formula.** The source notes print s = π·D_w(1 - σ)/(n_c·i), which makes spacing NARROWER as the wheel skids. A ground-driven wheel that skids turns FEWER times per metre travelled, so fewer seeds drop per metre and spacing gets WIDER — as the source's own MCQ (Q121), seeder manufacturers' manuals and the seed-drill calibration rule (Unit 3, Topic 1a: field rate is less than the stationary rate) all state. Dividing by (1 - σ) gives the correct direction: 5% skid → about 5.3% wider spacing.

$$\text{Plant population/ha} = 10^4/(s \times \text{row spacing})$$

Sizing starts from the agronomic population.

- **Vacuum disc design:** orifice dia ≈ **0.4–0.6 × seed dia**; vacuum **30–60 mbar** (higher for heavier seed); a singulator (scraper) removes doubles.

## Mass-flow metering (drills)

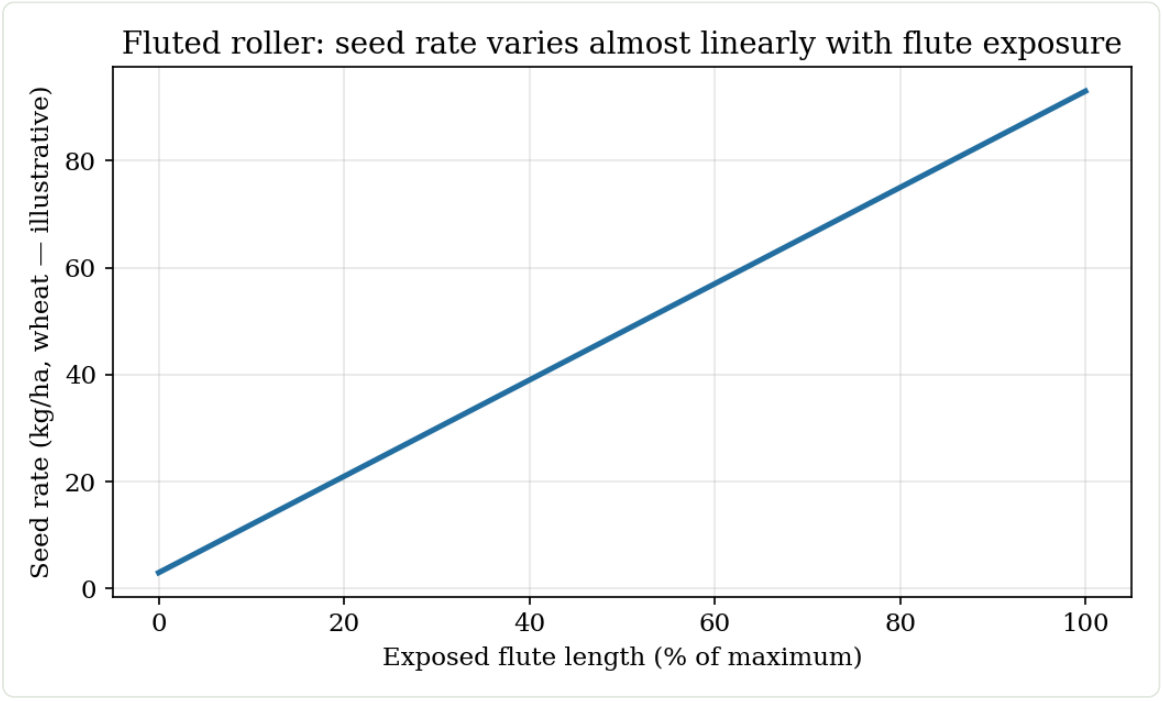



*Fluted roller metering*

- **Fluted roller (force feed):** seed rate ∝ exposed flute length (near-linear) and roller rpm. Size the flute shape (straight/helical — helical gives steadier flow), roller diameter and groove radius so seeds are not sheared at the cutoff; an adjustable spring-loaded gate protects large seeds.

$$Q\ (\mathrm{kg/s}) = \rho_b \times (a_f \times L_e \times z \times N/60) \times \eta_f$$

Bulk density × flute cross-section × exposed length × number of flutes × rpm × fill factor.




*Seed rate vs flute exposure*

- **Internal double-run:** two-diameter compartments for coarse/fine seeds; set the rate by speed ratio (gearing) rather than exposure.

- **Drive design:** ground wheel → chain/gear ratios; change the rate in steps with the gear ratio and finely with flute exposure.

## Practice MCQs

**Q111. The target cell fill for a precision planter plate is:**

(a) 50–60% (b) 70–80%
(c) 95–100% (d) Exactly 110%

**Q112. As plate (cell) linear speed increases, cell fill:**

(a) Increases (b) Decreases
(c) Constant (d) First falls then rises

**Q113. Barmington's data (Kepner Fig. 11.18) showed 100% cell fill for sugar-beet seed at a plate speed of about:**

(a) 10 fpm (b) 50 fpm
(c) 300 fpm (d) 500 fpm

**Q114. To plant at higher forward speeds with a plate meter, the designer should:**

(a) Speed up the plate (b) Use more cells per plate or a larger plate
(c) Reduce cells (d) Raise the hopper

**Q115. The metering-unit element that prevents multiple fill and seed breakage is the:**

(a) Knock-out (b) Cutoff device
(c) Agitator (d) Gate spring

**Q116. Positive removal of seed from cells is ensured by:**

(a) Gravity only (b) Knock-out/ejector devices
(c) Vacuum (d) Higher speed

**Q117. Conditions giving 95–100% cell fill also give:**

(a) Max damage (b) Most uniform spacing and minimum seed damage
(c) Higher draft (d) Poor spacing

**Q118. In fluted-roller metering, seed rate is chiefly controlled by:**

(a) Roller diameter (b) Exposed flute length
(c) Seed tube length (d) Hopper level

**Q119. Helical flutes (vs straight) on a fluted roller give:**

(a) Higher damage (b) Steadier, more uniform seed flow
(c) Lower capacity only (d) More wear

**Q120. Seed spacing delivered by a planter ($D_w$ wheel dia, $\sigma$ slip, $n_c$ cells, i plate/wheel ratio) is:**

(a) $\pi D_w(1-\sigma)/(n_c \cdot i)$ (b) $n_c \cdot i/\pi D_w$
(c) $\pi D_w \cdot n_c \cdot \sigma$ (d) $D_w/n_c$

**Q121. Ground-wheel slip (skid) of a drill causes actual seed spacing to be:**

(a) Narrower than theoretical (b) Wider than theoretical
(c) Unchanged (d) Random

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**Answers**

Q111 — (c) 95–100%
Q112 — (b) Decreases
Q113 — (b) 50 fpm

Q114 — (b) Use more cells per plate or a larger plate

Q115 — (b) Cutoff device

Q116 — (b) Knock-out/ejector devices

Q117 — (b) Most uniform spacing and minimum seed damage

Q118 — (b) Exposed flute length

Q119 — (b) Steadier, more uniform seed flow

Q120 — (a) $\pi D_w(1-\sigma)/(n_c \cdot i)$ — *Option (a) mirrors the source notes; the physically correct form divides by the skid term: $s = \pi D_w/[n_c \cdot i \cdot (1 - \sigma)]$ (Topic 6a).*

Q121 — (b) Wider than theoretical — *A skidding ground wheel turns fewer times per metre, so seeds fall farther apart than the theoretical $\pi D/(n_c \cdot i)$.*

**Sources:** Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 11 (Crop Planting), Ch. 12 (Row-Crop Cultivation), Ch. 13 (Application of Fertilizers); standard FMP design references.